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Ants, Plants, and Rats: Effects of Predator Exclusion on Chihuahua Plant Species Abundance

Introduction

Plant species abundance is a result of many abiotic and biotic factors, including predation. Two main plant predators of Chihuahua plants include ants and rodents.

We aimed to use existing, publicly available data by[1] to understand how each predator population (ants and rodents) influence plant species diversity. This data set collected longitudinal data plant abundance data, ant colony data, rodent data, and precipitation data from 1977-2002 across 24 experimental exclusion plots.

Materials and Methods

The data for this project is publicly available.

Results

Here is a table of which plots had which treatment(s).

```
library(knitr)
plots_treatments <- read.csv("./../data/data_raw/raw_data/plot_treatments.csv")
kable(plots_treatments, align = "cccc", caption = "Plot Treatments")
```

Table 1: Plot Treatments

Plot	X1977_1985	X1985_1987	X1988_2002
1	Mixed-size seeds added in pulse	All annuals removed	Banner-tailed kangaroo rat removed
2	Small seeds added	Summer annuals removed	Unmanipulated control
3	Kangaroo rats removed		
Pogonomyrmex rugosus removed		Kangaroo rats removed	

[1] S. K. Morgan Ernest, Thomas J. Valone, and James H. Brown, "Long-Term Monitoring and Experimental Manipulation of a Chihuahuan Desert Ecosystem Near Portal, Arizona, USA," *Ecology* 90, no. 6 (2009): 1708–8, https://doi.org/10.1890/08-1222.1.

Plot	X1977_1985	X1985_1987	X1988_2002
All ants removed			
4	All ants removed		
5	Banner-tailed kangaroo rats removed		All rodents removed
6	Large seeds added	Winter annuals removed	Kangaroo rats removed
7	All rodents removed		
8	Pogonomyrmex rugosus removed		All ants removed
9	Mixed-size seeds added	Bi-seasonal annuals removed	Banner-tailed kangaroo rats removed
10	All rodents removed		
All ants removed			
11	Unmanipulated control		
12	Pogonomyrmex rugosus removed		All ants removed
13	Large seeds added	Winter annuals removed	Kangaroo rats removed
All ants removed			
14	Unmanipulated control		
15	Kangaroo rats removed		
16	All rodents removed		
17	All ants removed		
18	Mixed-sized seeds added in pulse	All annuals removed	Kangaroo rats removed
19	Kangaroo rats removed		
Pogonomyrmex rugosus removed		Kangaroo rats removed	
All ants removed			
20	Mixed-sized seeds added	Biseasonal annuals removed	Kangaroo rats removed
All ants removed			
21	Kangaroo rats removed		
22	Small seeds added	Summer annuals removed	Unmanipulated control
23	All rodents removed		

Plot	X1977_1985	X1985_1987	X1988_2002
	All ants removed		
24*	Banner-tailed kangaroo rats removed		All rodents removed

Here is a figure of the linear mixed effects model and how it predicts the data.

Discussion

Conclusion

Bibliography

Ernest, S. K. Morgan, Thomas J. Valone, and James H. Brown. "Long-Term Monitoring and Experimental Manipulation of a Chihuahuan Desert Ecosystem Near Portal, Arizona, USA." *Ecology* 90, no. 6 (2009): 1708–8. https://doi.org/10.1890/08-1222.1.